

Phase 2 Data Architecture: Source Mapping Strategy

To build a comprehensive database of the life sciences ecosystem, we must combine regulatory registries (FDA) with financial, academic, and AI-driven data sources.

Below is the mapping of which data sources should be used to identify and classify each specific type of company.

Target Company Type	Primary Data Source(s)	Secondary / Enrichment Sources	Why this combination is required
Pharma & Therapeutics (Commercial / Manufacturing)	- FDA Drug Establishment Registration - FDA Biologics Registration	SEC EDGAR API	The FDA is the gold standard here. If a company is actively manufacturing a commercial drug or biologic, they <i>must</i> be registered with the FDA. SEC EDGAR can confirm financial status for public pharma.
Pharma & Therapeutics (Pure R&D / Pre-Clinical)	- NIH RePORTER API - USPTO PatentsView API	Biotech Directories (e.g., BioPharmGuy)	The FDA misses these entirely. Early-stage startups don't manufacture yet. To find them, you must track who is receiving NIH biomedical grants or filing medical patents.
CMOs & CDMOs (Contract Manufacturing)	- FDA Drug/Biologics Registration - SEC EDGAR API (10-K NLP)	OpenAI Web Search	FDA lists them, but doesn't label them. The FDA mixes internal pharma plants with external CDMOs. You <i>must</i> use OpenAI to scrape their sites to ask: <i>"Does this company accept external manufacturing contracts?"</i>
CROs (Clinical Research Organizations)	ClinicalTrials.gov API	- SEC EDGAR API - Biotech Directories - OpenAI Web Search	The FDA misses CROs entirely. The best way to find them programmatically is pulling the Sponsor/Collaborator fields from the ClinicalTrials.gov JSON API, then using OpenAI to verify their CRO status.

Summary of the Pipeline Strategy

- The Baseline (Manufacturing):** Download FDA Drug and Biologics registrations.

2. **The Clinical Layer (CROs):** Ingest ClinicalTrials.gov data to find clinical executors.
3. **The Innovation Layer (R&D):** Query NIH RePORTER and USPTO for pre-clinical startups.
4. **The AI Classifier (OpenAI):** Feed the raw list into your OpenAI web agent to explicitly classify the ambiguous FDA records (e.g., *Is this a private pharma manufacturer or a CDMO for hire?*).